STEP 8 OF 8

Ground Segment and Operations

Ground Station · Data Pipeline · Disaster Workflow · Mission Operations — FINAL

Colombo Primary GS | SatNOGS Backup | NDMSC Integration | Automated Pipeline | < 35 min Alert Latency | Open Data Policy

8

OFFICIAL MISSION PARTNER CO-DESIGN DOCUMENT

MTQ-Only Revision | Reaction Wheel Removed | 3-Axis Magnetorquer System

1 Ground Segment Overview — Closing the Loop

The ground segment is where SLARS-BCG-1 fulfils its purpose. Every engineering decision in Steps 1 through 7 — orbit, power, payload, COMMS, ADCS, OBSW, structure — all exists to produce data that flows through the ground segment into the hands of Sri Lanka's disaster management community. MTQ-Only changes to Step 8 are limited: RW anomaly procedures are removed; the science gate threshold is updated; and the TC count is 30.

Colombo GS	SatNOGS	< 35 min	4 types	30 TCs
Primary station	Global UHF backup	Flood alert latency	Science data products	Command set

2 Primary Ground Station — Colombo, Sri Lanka

Component	Specification	Cost (USD)	Supplier Options
UHF Yagi antenna	10-element, ~10 dBi, 437 MHz, circular pol.	200–500	M2 Antennas, Arrow, custom
VHF Yagi antenna	6-element, ~8 dBi, 145 MHz, for APRS	150–350	M2 Antennas, amateur suppliers
S-band parabolic dish	1.0 m diameter, offset-fed, ~25 dBi at 2.4 GHz	600–1,800	Modified WiFi dish, SAT-Service
S-band feed + LNA	Custom feed horn + LNA NF < 1.5 dB at 2.4 GHz	200–500	Kuhne Electronic, Down East
Az-El rotator	Computer-controlled, 360 az / 0-90 el	600–2,000	Yaesu G-5500, SPID RAK/RAS
UHF/VHF SDR receiver	RTL-SDR v3 with LNA4ALL for UHF telemetry	50–150	RTL-SDR.com, Nooelec
S-band SDR transceiver	USRP B210 or LimeSDR for 128 kbps S-band DL	1,000–2,500	Ettus Research, Lime
Ground station PC	Ubuntu Linux, Intel i7, 16 GB RAM, 2 TB SSD	600–1,500	Local procurement
UPS power backup	2 kVA, 4-hour battery runtime for disaster ops	500–900	APC or Schneider
Total estimated cost	**Complete primary ground station**	**USD 3,900–10,200**	**University budget achievable**

3 Ground Station Software Stack

Software	Function	License	Notes
GPredict	Satellite tracking, pass prediction, rotator control	Free / GPL	Load TLE from Celestrak daily
GNURadio 3.10	SDR signal processing — UHF demod + S-band CFDP	Free / GPL	Custom flowgraphs GMSK + BPSK
gr-satellites	Pre-built GNURadio blocks for CSP/AX.25 CubeSat	TMFree / GPL	GMSK/AX.25 telemetry decode
SatNOGS client	Automated pass scheduling, upload to SatNOGS network	Free / AGPL	Runs as service
Custom TC/TM software	Send all 30 TCs; parse all TM fields; FDIR alerts	Team-developed Python	~3,000 lines; Django web UI
CFDP engine	CCSDS File Delivery Protocol for S-band imagery	libcfdp open source	Handles retransmission + CRC

Data pipeline (Python)	JPEG2000 decode, radiometric calibration, NDVI	Team-developed	GDAL + numpy + rasterio
PostGIS database	Store imagery, AIS tracks, APRS messages, TM log	Free / PostgreSQL	Persistent archive; REST API
Web dashboard (Django)	Real-time status, pass schedule, product viewer	Team-developed	Accessible to NDMSC via browser
Alert system	Email/SMS/API alert to NDMSC when flood threshold met	Team-developed	NDVI delta or APRS flood message

ALL GROUND STATION SOFTWARE IS FREE AND OPEN SOURCE — ZERO LICENSING COST.
Custom development required: TC/TM app (~3,000 lines Python), data pipeline (~1,500 lines), web dashboard (~2,000 lines).
Achievable by 3-4 software engineering students over 6-9 months.

4 Automated Data Processing Pipeline

Stage	Input	Process	Output	Latency
Frame sync + CR	CRaw CFDP bits from S-band	SDRSync CCSDS frames; verify CRC16; discard corrupted	Validated CFDP data	< 1 min
CFDP reassembly	Validated frames	Reassemble file from packets; request retransmit if gaps	JPEG2000 image files	< 2 min
Image decode	JPEG2000 compressed files	OpenJPEG lossless decode; restore to 16-bit	TIFF per band5-band TIFF stack	< 3 min
Radiometric cal.	Raw DN values	Apply pre-launch calibration coefficients; DN to radiance	Calibrated radiance TIFF	< 5 min
Geometric corr.	Calibrated TIFF + ADCS	Altitude + TLEOrthorectification using SRTM DEM; project to WGS84	Geo-registered GeoTIFF	< 15 min
NDVI computation	Red + NIR band GeoTIFFs	Pixel NDVI = (NIR-Red)/(NIR+Red); apply cloud mask	NDVI GeoTIFF + change	map< 20 min
Flood detection	NDVI map + NIR band	Threshold NDVI < -0.05 as water; compare to baselin	Flood polygon (GeoJSN)	< 25 min
Archive + alert	All derived products	Write to PostGIS; thumbnails; update dashboard	Published + NDMSC alert	< 35 min

5 Disaster Response Workflow

Scenario	Timescale	SLARS-BCG-1 Contribution	Data Product
Major flood — Kelani River basin	T+0 to T+6 hr	Imaging pass captures inundation extent; NDVI change map	Flood extent GeoJSON polygon
Cyclone landfall — coastal damage	T+0 to T+12 hr	SWIR captures coast before and after; change detection	Coastline shift map
Drought — North Sri Lanka	Ongoing, weekly	NDVI time series every overpass; vegetation stress ind	Crop health index by districtx
Tsunami aftermath — coastal	surveyT+1 day	Multispectral survey of coastal strip; pre/post compare	Before/after mosaic
Field team lost comms — rescue	T+0 (immediate)	Team activates APRS terminal; GPS + status uplinked	GPS relayed to NDMSC next pass
Port disruption assessment	T+0 to T+5 hr	AIS vessel traffic pattern change detects disruption	Affected port identified auto.

6 Anomaly Response Procedures — MTQ-Only Updated

Anomaly	Indicators	Response Procedure	TC Commands
Satellite not heard	No beacon 3+ passes	Check SatNOGS; verify GS; wait 24 hr	—

Battery low SoC < 25%	TM: SoC field < 0.25	SET_MODE SAFE; suspend science; monitor	0x01, 0x09
ADCS anomaly — tumbling	TM: rate > 5 deg/s	SET_ADCS_MODE 1 (B-dot); wait 3 orbits	0x0A
MTQ not responding	TM: ADCS error > 14 deg persistent	SET_ADCS_MODE 1; RESET_SUBSYSTEM ADCS rail	0x0A, 0x04

FDIR Level 4 alert	TM: mode = EMERGENCY	Read FDIR log; diagnose; issue corrective	TC0x11, 0x04, 0x01
S-band TX temp > 60 C	TM: tx_temp > 60	Reduce science to 1/day; monitor recovery	0x05 n=1
eMMC flash full	TM: flash_usage > 95%	FORMAT_FLASH (last resort) or DL + delete	0x0D
TLE stale (> 5 days)	Beacon timing drift	Upload fresh TLE immediately	0x02
No S-band downlink	UHF OK but no imagery	Check CFDP; restart via extended TM request	0x07, 0x0C
Imager not responding	Science data absent	Disable imager; power cycle rail; re-enable	0x04, 0x08, 0x09
REMOVED: RW saturated	—	***Reaction wheel not present — procedure***	***deletedN/A***
REMOVED: RW desat TC	—	***TC 0x14 removed — no momentum to dump***	***N/A***

7 SatNOGS Global Backup Network

Action	Detail	Timeline
Register satellite	Create entry on db.satnogs.org with callsign, frequencies, TLE	6 months before launch
Define TM decoder	Write gr-satellites decoder for CSP/AX.25 TM; upload to SatNOGS	4 months before launch
IARU coordination	437.550 MHz confirmed by IARU before SatNOGS listing	12 months before launch
Set observation priority	Mark SLARS-BCG-1 high-priority in SatNOGS scheduler	At TLE release post-launch
Download backup TM	SatNOGS API provides all received frames if primary GS misses pass	Automated — every 2 hours
Acknowledge community	Credit SatNOGS contributors in all publications	Ongoing

8 Launch Preparation Milestones

Milestone	Responsible	Target Timing	Dependency
IARU frequency coordination (437 + 145 MHz)	Mission team via RSSL	18 months before launch	Before launch provider contract
TRCSL frequency coord (2.4 GHz S-band)	Mission team via TRCSL	18 months before launch	Before launch provider contract
ITU satellite network filing	TRCSL files on behalf of Sri Lanka	18 months before launch	After TRCSL coordination
Select launch provider	Mission team	12 months before launch	After CDR; mass / size confirmed
Interface Control Document to deployer	Mission team to launch provider	9 months before launch	After Step 7 structure finalised
Vibration qualification test	Mission team at test facility	8 months before launch	After full AIT
TVAC test	Mission team at vacuum facility	7 months before launch	After vibration test

Delivery to launch site	Mission team courier + escort	3 months before launch	After FAR; IATA A803 shipping
RF quiet test inside deployer	Mission team + launch provider	2 months before launch	Spectrum analyser confirms no RF

9 Complete Mission Summary — All 8 Steps (MTQ-Only Final)

Step	Title	Key Output	Status
1	Orbit Design	500 km LEO, 10 deg incl., 5-6 passes/day over Sri Lanka	LOCKED
2	Power Budget (EPS)	12 Wh battery, ~2.0 W avg, 3 science passes/day	LOCKED
3	Payload Stack	5-band imager + SDR relay + AIS, 67 mm, 2.15 W peak	LOCKED
4	COMMS Subsystem	UHF 9.6 kbps + S-band 128 kbps + APRS — all links +margin	LOCKED
5	ADCS — MTQ Only	3-axis MTQ, B-dot/PD/PID, +/-6-8 deg nadir pointing, no RW	LOCKED
6	OBC and Flight SW	ARM Cortex-M4, FreeRTOS 7 tasks, 30 TCs, 4-level FDIR	LOCKED
7	Structural and Thermal	Al 6061-T6, 2,445 g (215 g margin, 8.1%), passive TCS	LOCKED
8	Ground Segment + Ops	Colombo GS, pipeline, NDMSC, SatNOGS, MTQ procedures	LOCKED

Item	Result (MTQ-Only Final)
All 8 design steps	Complete — mission fully designed, magnetorquer-only ADCS
Orbit selected	500 km LEO, 10 deg inclination, 5-6 passes/day over Sri Lanka
Power budget	~2.0 W avg, 12 Wh battery, 3 science passes/day
Payload instruments	3 instruments: multispectral imager + SDR relay + AIS receiver
Communication links	4 links, all positive margin: UHF DL/UL + S-band + APRS
ADCS control	3-axis MTQ only: B-dot + PD + PID, +/-6 to 8 deg nadir pointing
Flight software	7 FreeRTOS tasks, 30 telecommands, 4-level FDIR, ARM Cortex-M4
Spacecraft mass	2,445 g of 2,660 g maximum — 215 g margin (8.1%)
Disaster alert latency	Flood map to NDMSC within 35 minutes of each pass
Mission life	2+ years nominal operations; passive deorbit ~4 years post-launch
Reaction wheel	**REMOVED — 3-axis magnetorquers only; no moving ADCS components**

ALL 8 STEPS LOCKED — MTQ-ONLY MISSION DESIGN COMPLETE. Proceed to build, integration, test, and launch.

THE PURPOSE OF SLARS-BCG-1 — FULFILLED

"SLARS-BCG-1 is a satellite created by Sri Lankan students to strengthen our island's resilience against disasters. From orbit, it observes floods, coastline shifts, and vegetation stress, turning environmental signs into knowledge. It supports early warnings — ensuring vital information reaches people even when local networks fail. More than a scientific mission, it is a symbol of youth innovation and national hope."